COS-AT2-Pseudocode Writing

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Course : CSA0712 - Computer Network.

Question 1: Write a Pseudocode function SecureSendEmail (Sender, recipient, messageBody) that contains to an SMTP Server on port 25. Your logic must inspect the 3-digit numeric Status code returned by the server after a single command (HELLO, MAIL FROM, RCPT TO, DATA) If any command returns a failure status code (4xx or 5xx) the pseudocode must immediately send QUIT, abort the operation and return a clear error message detailing which step failed.

Answer:

```
FUNCTION SecureSendEmail (sender, recipient, messageBody)
    CONNECT to SMTP Server on port 25

    SEND "HELLO Client.local"
    response ← Read_message()
    IF response.code >= 400 THEN
        SEND "QUIT"
        CLOSE connection
        RETURN "Error: HELLO failed - Code " + response.code
    END IF

    SEND "MAIL FROM: <" + sender + ">"
    response ← READ_RESPONSE()
    IF response.code >= 400 THEN
        SEND "QUIT"
```

```
        CLOSE  connection
        Return "Error: MAIL FROM failed - code "+response.code
    END IF

SEND "RCPT TO:<" + recipient +">"
response ← READ_RESPONSE()
IF response.code >= 400 THEN
    SEND "QUIT"
    CLOSE connection
    RETURN "Error: RCPT TO failed - code" + response.code
END IF

SEND "DATA"
response ← READ_RESPONSE()
IF response.code >= 400 THEN
    SEND "QUIT"
    CLOSE connection
    RETURN "Error: DATA failed - code" + response.code
END IF

SEND messageBody
SEND "."
response ← READ. RESPONSE()
If response.code >= 400 THEN
    SEND "QUIT"
    CLOSE connection
    RETURN "Error: MESsage trasmission failed- code" + code.
END IF

SEND "QUIT"
CLOSE connection

RETURN "Email sent successfully"
```

Question 2: Write a Pseudocode Procedure SyncInboxPOP3 (usernamp, password, localDatabase) that connects to a POP3 server on Port 110. The Procedure should log in and check the total no.of message using STAT. Loop through each message, but instead of downloading blindly, use the UIDL command to fetch the unique identifier of each email. Check this identifier against the local Database.

```
PROCEDURE SyncInboxPOP3 (username, password, localDatabase)

    CONNECT to POP3 Server on Port 110

    SEND "USER" + username
    CHECK response.

    SEND "PASS" + password
    CHECK response.

    SEND "START"
    response ← READ_RESPONSE()
    totalmessages ← EXTRACT_MESSAGE_COUNT (response)

    FOR messagenumber ← 1 TO totalMessages

        SEND "UIDL" + messagenumber
        response ← READ_RESPONSE()
        uniqueID ← EXTRACT_UID (Response)

        IF localDatabase.CONTAINS (uniqueID) THEN
            CONTINUE
        END IF

        SEND "RETR" + messageNumber
        response ← READ_RESPONSE()
```

```
        email ← READ. COMPLETE_ MESSAGE ( )
        localDatabase. SAVE (uniqueID, email)
    END FOR

    SEND "QUIT"
    CLOSE connection

END PROCEDURE
```

Question 3: Write a Pseudocode script that connects to an IMAP server on port 143 to achieve old emails. The Script must log in, select the INBox folder and search for all read message to an archive folder named OldMail using the COPY command. Once successfully copied, the code must add the Deleted flag to the original message

```
CONNECT to IMAP server on port 143
SEND "A001 LOGIN username password"
SEND "A002 SELECT INBox"
SEND "A003 SEARCH SEEN"
response ← READ_RESPONSE ( )
messageIDS ← EXTRACT_MESSAGE_IDS (response)
FOR EACH messageID IN messageIDs
    SEND "A004 COPY " + messageID + " OldMail"
    response ← Read_ RESPONSE ( )
```

```
        IF response indicates success THEN
            SEND "A005 STORE" + messageID + " +FLAGS (\Deleted)"
            response ← READ_RESPONSE()
        ENDIF

    END FOR

    SEND "A006 EXPUNGE"

    SEND "A007 LOGOUT"

    CLOSE connection.
```

Question 4.

Write a Pseudocode function Robust NetWord Reader (socket, timeout_limit) designed to safely read data streams from an active mail client socket. The logic must track the time taken for a server to respond within timeout limit seconds, if must trigger a network retry mechanism up to 3 times. If all.

```
FUNCTION Robust NetWork Reader (Socket, timeout limit)

    maxRetries ← 3
    attempt ← 0

    WHILE attempt < maxRetries

        attempt ← attempt + 1
        startTime ← CURRENT_TIME()

        SET_SOCKET_TIMEOUT (socket, timeout limit)

    TRY
```

```
        data ← READ_FROM_SOCKET (socket)

        endTime ← CURRENT_TIME()

        IF (endTime - startTime) <= timeoutLimit THEN
            RETURN data
        END IF

    CATCH TimeoutException

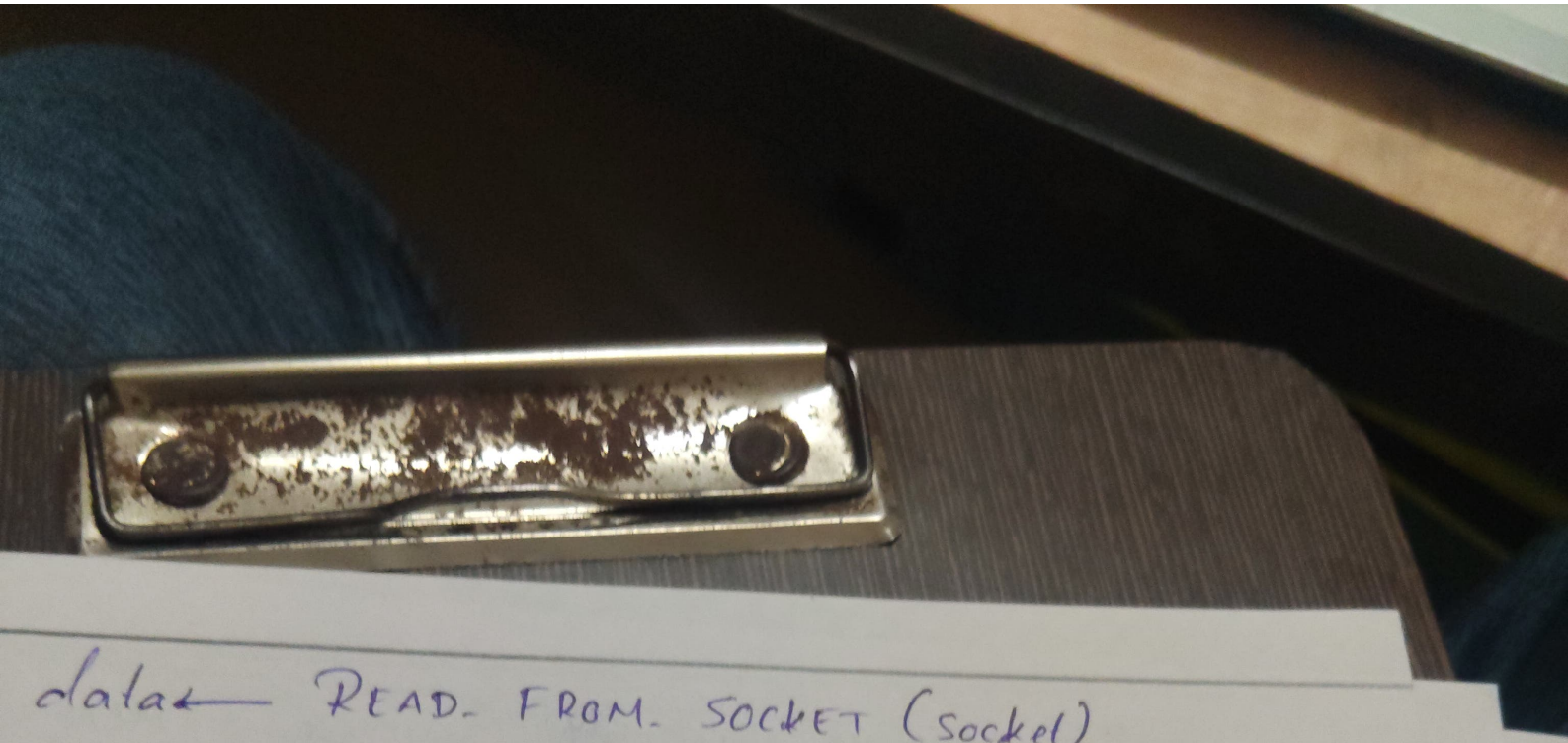
        If attempt < maxRetries THEN
            RETRY

        ELSE
            FORCE_CLOSE (socket)
            THROW ConnectionTimeoutException
        END IF

    END TRY

  END WHILE

END FUNCTION.
```



data ← READ_FROM_SOCKET (socket)

endTime ← CURRENT_TIME()

IF (endTime - startTime) <= timeoutLimit THEN
    RETURN data
END IF

Timeout Exception

f attempt < maxRetries THEN
  RETRY

SE
  FORCE_CLOSE (socket)
  THROW Connection Timeout Exception
D IF